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5 August 1981
SUPERSEDING
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23 April 1976

MILITARY SPECIFICATION

STEREOSCOPES, LENS, AERIAL PHOTOGRAPH INTERPRETATION (POCKET)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two- and four-power magnifying-lens pocket stereoscopes.

1.2 Classification. The stereoscopes shall be of the following types, as specified (see 6.2):

Type I - 2 power.
Type II - 4 power.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
QQ-W-423	- Wire, Steel, Corrosion-Resisting.
CCC-C-700	- Cloth, Coated, Vinyl Coated (Artificial Leather).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-150	- Photographic Lenses.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASTM A 229 - Steel Wire, Oil-Tempered for Mechanical Springs.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

NATIONAL BUREAU OF STANDARDS (NBS)

NBS 374 - Methods for Determining the Resolving Power of Photographic Lenses.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The stereoscopes shall consist of two lenses mounted in a frame supported by folding legs.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3 Frame. The frame shall be aluminum, zinc, or magnesium. No component of the stereoscope shall extend above the frame except for the upper edge of the lens barrels. The overall dimensions of the assembled folded stereoscope shall be not more than 5.75 inches long x 3 inches wide x 1 inch thick.

3.3.1 Interpupillary adjustment. Provision shall be made to accommodate interpupillary distances between 55 and 72 millimeters (mm). A positive means for locking in any position shall be provided. The interpupillary adjustment shall be provided with index marks at 1-mm intervals to facilitate presetting the instrument. The method of adjustment shall provide positive, simple, and easy adjustment, except that, when specified (see 6.2), a positive screw adjustment with a knurled head shall be provided. At any interpupillary setting, no part of the instrument shall obstruct or interfere with the nose of the operator.

3.4 Legs. The legs shall be zinc-coated steel wire conforming to ASTM A 229 or corrosion-resisting steel wire conforming to QQ-W-423. The legs shall support the stereoscope at the optimum viewing distance. The portion of the legs which will contact the photographs or other documents shall be rounded, beveled, or formed so as to prevent tearing, scratching, or otherwise mutilating the surface on which it rests. Positive stops, notches, or springs shall hold the legs in both open and closed positions. When set up for viewing at any interpupillary setting, the stereoscope shall stand firm without wobbling.

3.5 Lenses. The lenses shall be precision ground, with spherical and chromatic aberrations corrected. The lenses shall be free of striae, bubbles, scratches, or other imperfections and shall show no perceptible distortion throughout their respective fields. The focal lengths of the lenses shall not exceed 1 percent of each other. The left and right lenses shall be replaceable and interchangeable. The lenses shall not be cemented, spun, or burnished into the frame.

3.6 Finish. Unless otherwise specified (see 6.2), the stereoscope finish shall be at the contractor's option.

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3.7 Case. A case to protect the stereoscope when not in use shall be provided. The case shall be artificial leather conforming to CCC-C-700, class optional, Treatment b, or soft, flexible plastic. The material shall be not less than 0.047 inch thick. The case shall have a flap with a snap fastener to keep the cover closed. The snap fastener shall be positioned so that it will not contact the lens surface.

3.8 Identification marking. The stereoscope and the case shall be identified in accordance with MIL-STD-130.

3.9 Type I, two-power. The lenses shall be two-element achromatic. Each lens shall cover a flat surface not less than 90 mm in diameter and shall resolve not less than 16 lines per millimeter at the center of the field of view and not less than 10 lines per millimeter at the edge. Each lens shall provide a magnification of between 1.9 and 2.1 diameters. The lenses shall be held in place by spring clamp rings or threaded cells.

3.10 Type II, four-power. The lenses shall be two-element achromats or three-element apochromats, at the contractor's option. Each lens shall cover a flat surface of not less than 50 mm in diameter and shall resolve not less than 20 lines per millimeter at the center of the field of view and not less than 12 lines per millimeter at the edge. Each lens shall provide a magnification of between 3.8 and 4.2 diameters.

3.10.1 Mounts. The lenses shall be mounted by spring clamp rings or threaded cells, at the contractor's option. If threaded cells are selected, the cells shall be threaded to provide a vertical adjustment of not less than 2.50 mm above and below the focal point.

3.11 Workmanship. All surfaces shall be smooth and free from warp, bow, twist, pits, scales, and any other defect that may affect appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Inspection Level S-2.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. Stereoscopes shall be examined as specified herein for the following defects:

Major

101. Materials not as specified.
102. Frame not as specified.
103. Interpupillary adjustment not as specified.
104. Legs not as specified.
105. Lenses not as specified.
106. Mounts not as specified.
107. Workmanship not as specified.

Minor

201. Case not as specified.
202. Identification marking missing, incomplete, or illegible.

4.4.2 Tests.

4.4.2.1 Resolving power. The stereoscope shall be set up over a resolving-power test target. The resolving-power test targets specified in MIL-STD-150 and in NBS 374 are acceptable for this test. A person with normal 20/20 vision shall view the target through the center and at least two places on the side of each lens individually. Nonconformance to 3.9 or 3.10 shall constitute failure of this test.

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4.4.2.2 Focal length. The focal length shall be determined by placing the lens to be tested parallel to a plane surface at such a distance as to achieve the best possible axial focus of a light source which must be at least 20 feet from the lens unless a collimator and reticle are used. The distance between the centerline of the lens and the image on the plane surface shall be measured to the nearest 0.016 inch. This is the focal length. Any standard optical method giving a greater degree of accuracy may be substituted for this test. The focal length of both lenses of each stereoscope being tested shall be determined and compared. Any difference more than 1 percent shall constitute failure of this test.

4.4.2.3 Magnification. The magnification is the quotient of 10 divided by the focal length (see 4.4.2.2). Nonconformance to 3.9 or 3.10 shall constitute failure of this test.

4.4.2.4 Chromatic aberration. A high-contrast, white test pattern with a black background illuminated by natural sunlight, or any artificial means having the full spectral range of natural sunlight, shall be viewed through the stereoscope by a person having full color vision. Any evidence of color fringes shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Materials, methods, or containers not as specified.	Each incorrect material, method, or container shall constitute one defect.
109.	Quantity per pack not as specified.	
110.	Strapping not as specified.	
111.	Marking missing, illegible, incomplete, or incorrect.	

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each stereoscope, in its case, shall be preserved in accordance with MIL-P-116, Method IA-8.

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5.1.1.1 Consolidation. Stereoscopes preserved as specified in 5.1.1 shall consolidated in close-fitting boxes conforming to PPP-B-636, W6c, style optional, not to exceed 50 per pack.

5.1.2 Commercial. Stereoscopes shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. The stereoscopes shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional, or PPP-B-621, Class 2, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Packing shall be as specified for Level A, except that boxes shall be Domestic Type or Class and strapping shall be Finish A.

5.2.3 Commercial. Stereoscopes shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Levels A and B. Level A and B marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The pocket stereoscope is intended to be used for the stereoscopic examination of aerial photographs.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of stereoscope required (see 1.2).
(c) When a positive screw adjustment with a knurled head shall be provided (see 3.3.1).
(d) When the finish shall not be at the option of the contractor (see 3.6).
(e) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(f) When strapping shall be other than as specified (see 5.2.1).

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6.3 Definitions and nomenclature. Definitions and nomenclature of terms used in this specification may be found in MIL-STD-150 and MIL-STD-1241.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

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Review activity:
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User activities:
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